Project: Worked examples
Title: Heat sink thermal resistance chain
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the steady-state source temperature from a series chain of thermal resistances to ambient, with the rise across each stage. It gives the margin against the permitted maximum, negative over the limit, and the largest total resistance allowed.

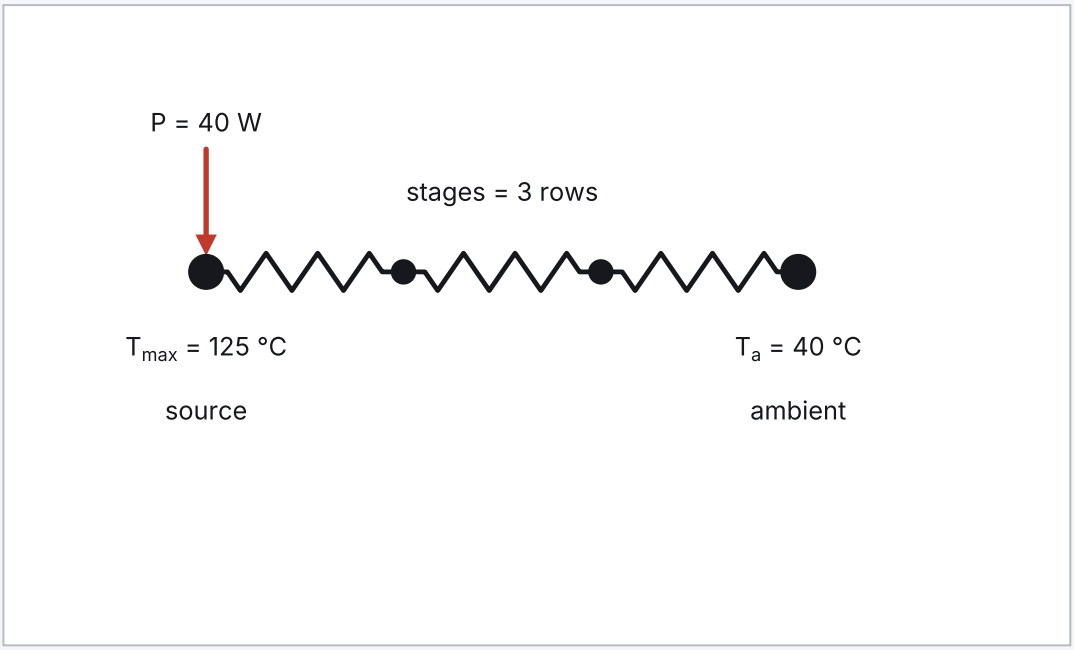


Figure 1 — Thermal resistances in series from the source to ambient, hottest first (three drawn; the stages as listed). Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Temperature rise across each stage	$\Delta T_i = P \cdot R_i$
Total resistance of the chain and the total rise	$R_{tot} = \Sigma\, R_i$; $\Delta T_{tot} = P \cdot R_{tot}$
Source (hottest node) temperature, each node below it being the one above less that stage's rise	$T_1 = T_a + \Delta T_{tot}$
Margin against the permitted maximum	margin = $T_{max} - T_1$
Largest total resistance the limit allows, the heat sink budget	$R_{max} = (T_{max} - T_a)/P$
Utilisation of the permitted rise	$u = \Delta T_{tot}/(T_{max} - T_a)$

Basis: Integrated circuit thermal measurement method: electrical test method (junction-to-reference resistances)[1]; Thermal resistance networks in the cooling of electronic equipment[2](§15-2, 5th ed. (edition to be checked)); Thermal resistance in series[3](§3.1, 7th ed. (edition to be checked)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Steady power dissipated (the heat flow through every stage)	40	W	spec: DS-PWR-902 device and heat sink datasheets rev A	0 … 10000 W
T_a	Ambient the last stage rejects to	40	°C	spec: SPEC-EL-05 thermal limits	-40 … 85 °C
stages	Thermal resistances in series, hottest first	3 row(s)		spec: DS-PWR-902 device and heat sink datasheets rev A	any
T_{max}	Maximum permitted temperature of the hottest node (junction or case)	125	°C	spec: SPEC-EL-05 thermal limits	40 … 200 °C

TABLE ROW	VALUES
stages[1]	name = junction to case; R = 0.5 K/W
stages[2]	name = case to sink; R = 0.2 K/W
stages[3]	name = sink to ambient; R = 1.8 K/W

3. Assumptions

A1. Steady state: the whole dissipated power flows through the one chain, and each resistance is constant. (No thermal capacitance (nothing about warm-up or a transient overload); no parallel path through the board.)
A2. The stated ambient is the temperature the last stage actually rejects to. (Air preheated by neighbouring parts can be well above room temperature.)

4. Working

[4] $R_{tot} = \Sigma R_i$ [3](§3.1)
= 0.5 + 0.2 + 1.8 K/W
= 2.5 K/W
Total thermal resistance
[5] $dT_{tot} = P * R_{tot}$
= 40 W × 2.5 K/W
= 100 K
Total rise from ambient to the source
[6] $T_1 = T_a + dT_{tot}$ [2](§15-2)
= 40 degC + 100 K
= 140 degC
Source temperature (hot side of junction to case)
[7] $dT_1 = P \cdot R_1$
= 40 W × 0.5 K/W
= 20 K
Rise across junction to case
[8] $dT_2 = P \cdot R_2$
= 40 W × 0.2 K/W
= 8 K
Rise across case to sink
[9] $T_2 = T_1 - dT_1$
= 140 °C − 20 K
= 120 degC

Hot side of case to sink

[10] $dT_3 = P \cdot R_3$

$= 40\ W \times 1.8\ K/W$

$= 72\ K$

Rise across sink to ambient

[11] $T_3 = T_2 - dT_2$

$= 120\ °C - 8\ K$

$= 112\ degC$

Hot side of sink to ambient

[12] $dT_m = T_{max} - T_1$

$= 125\ degC - 140\ degC$

$= -15\ K$

Thermal margin (negative when over the limit)

[13] $dT_{allow} = T_{max} - T_a$

$= 125\ degC - 40\ degC$

$= 85\ K$

Permitted rise from ambient to the source

[14] $u = dT_{tot} / dT_{allow}$

$= 100\ K / 85\ K$

$= 1.1765$

Utilisation: rise over permitted rise

[15] $R_{max} = (T_{max} - T_a) / P$

$= (125\ degC - 40\ degC) / 40\ W$

$= 2.125\ K/W$

Largest total resistance the limit allows (heat sink budget)

5. Results

SYMBOL	VALUE	DESCRIPTION
T_1	140 °C	Source temperature
dT_m	-15 K	Thermal margin
u	1.18	Utilisation
R_{max}	2.125 K/W	Heat sink budget
R_{tot}	2.500 K/W	Total thermal resistance
dT_{tot}	100 K	Total rise (governing)
dT_1	20 K	Rise across stage 1
dT_2	8 K	Rise across stage 2
T_2	120 °C	Hot side of stage 2
dT_3	72 K	Rise across stage 3
T_3	112 °C	Hot side of stage 3

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Rise from ambient against the permitted rise	100 K	85 K	1.18	[2](§15-2)

7. Applicability

None.

8. Verification

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Template elec.thermal-chain@1.1.1 Content hash a7184565e228e0e8 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. JEDEC Solid State Technology Association. Integrated Circuits Thermal Measurement Method—Electrical Test Method (Single Semiconductor Device). JESD51-1. JEDEC.
2. Çengel YA, Ghajar AJ. Heat and Mass Transfer: Fundamentals and Applications. 5th ed. McGraw-Hill Education; 2015.
3. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.